Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Rails: VBUS -> VSYS 3.0–4.5 V (LED SW lines) -> +3V3 -> +1V1 at the RP2350.
Nets between sheets are global labels; rails are power symbols.
Requirements: design/intent.md. Parts: design/parts/. Decisions: design/decisions.md.

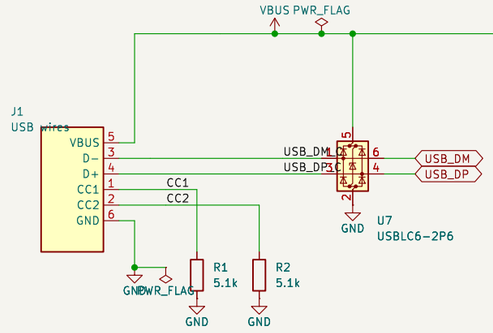
Drawn by scripts/schematic.py
234 RGB LEDs on a 38 mm round board
CERN-OHL-P-2.0

Sheet: /
File: round-pendant.kicad_sch

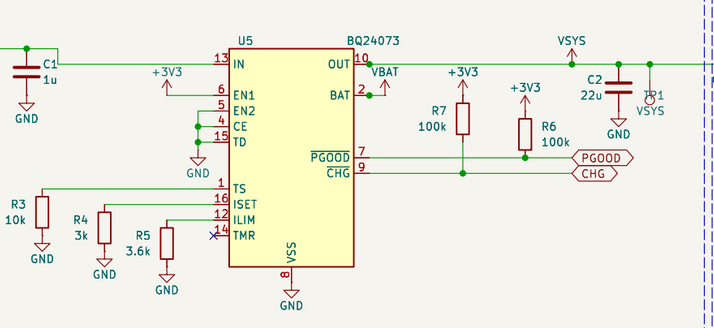
Title: round-pendant: top

Size: A4	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/5

USB WIRES, ESD

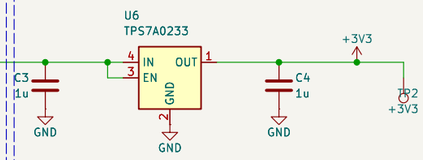


CHARGER AND POWER PATH



USB500: 450−500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

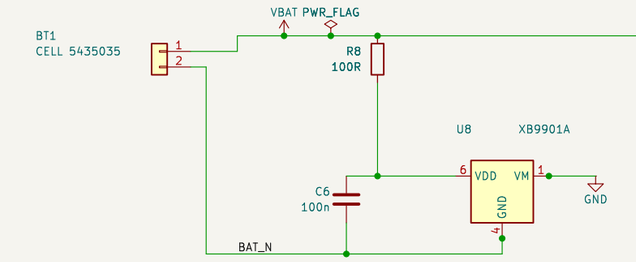
3V3



3V3 only ever follows VSYS down: the driver inputs
stay within VCC + 0.3 V (drivers.md). 150 mA max.

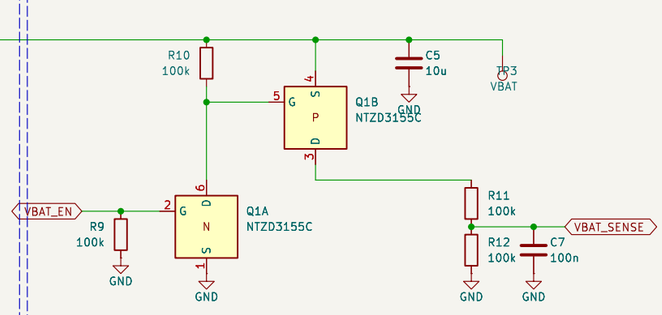
CELL AND PROTECTION

Hand−soldered leads. 610 mAh listed, charged to 4.2 V (battery.md)



BAT_N: cell minus. Pins 4, 5 and the pad. Never pour GND here,
or protection is bypassed (protection.md).

VBAT SENSE



Zero draw off. ADC reads true only while 3V3 regulates.
VBAT >= 3.5 V (intent R22). Wait >= 5 ms after enabling.

Drawn by scripts/schematic.py
USB−C, charger, protection, LDO, VBAT sense
CERN−OHL−P−2.0

Sheet: /Power/
File: power.kicad_sch

Title: round−pendant: power

Size: A3	Date: 2026−11−02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 2/5

RP2350A, FLASH, CRYSTAL

Regulator: C16 CIN, C18 COUT, L1's polarity and copper follow chips/rp2350.md

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C8–C12), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C13), ADC_AVDD 44 and IOVDD 45 (C14).

C18 COUT at L1; C19 at DVDD 23; C20, C21 at DVDD 6, 39

I2C0 to both drivers, 1 MHz: 12 mA drive, LGC = 1 (drivers.md). Unused GPIOs are left open; firmware clears latched pull-downs before sleep on GPIO 2–5, 8, 9, 12, 13, 26.

Flash pad: float (no internal connection)

VREG_PGND joins GND at one point

ACCELEROMETER (WAKE)

Address 0x19. CS, SAO tied high: ~20 k internal pull-ups would draw 160 uA if strapped low.

BUTTONS

Internal pull-up; wake on low. Hold SW2, tap SW1 for BOOTSEL.

I2C0 PULL-UPS

TEST PADS

CENTRE LED

Average about 1.5 mA per die at full, near a matrix LED's 1.25 mA. Firmware: GPIO high is off; clear pull-downs before sleep.

Drawn by scripts/schematic.py
RP2350A, flash, crystal, accelerometer, buttons
CERN-OHL-P-2.0
Sheet: /MCU/
File: mcu.kicad_sch

Title: round-pendant: mcu

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 3/5

DRIVER 1

Caps at PVCC 15 and 25 and AVCC 14; ground to the chip's GND.
22 uF bulk: edges only. 494 mA scan peak (drivers.md).

VSYS
C30 22u
C31 1u
C32 100n
C33 1u
C34 100n
C35 1u
C36 100n
+3V3
R30 100k
U1
IS31FL3741A
I2C_SDA
I2C_SCL
INTB1
SDB1
C37 100n
R31 100k
R32 30k
ADDR GND: 0x30
30 k: 12.67 mA peak,
1.25 mA average per die

9	SDA	SW1	16 SW1_1
10	SCL	SW2	17 SW1_2
7	INTB	SW3	18 SW1_3
11	SDB	SW4	19 SW1_4
		SW5	20 SW1_5
		SW6	21 SW1_6
13	ISET	SW7	22 SW1_7
8	ADDR	SW8	23 SW1_8
		SW9	24 SW1_9
		CS1	26 CS1_1
		CS2	27 CS1_2
		CS3	28 CS1_3
		CS4	29 CS1_4
		CS5	30 CS1_5
		CS6	31 CS1_6
		CS7	32 CS1_7
		CS8	33 CS1_8
		CS9	34 CS1_9
		CS10	35 CS1_10
		CS11	37 CS1_11
		CS12	38 CS1_12
		CS13	39 CS1_13
		CS14	40 CS1_14
		CS15	41 CS1_15
		CS16	42 CS1_16
		CS17	43 CS1_17
		CS18	44 CS1_18
		CS19	45 CS1_19
		CS20	46 CS1_20
		CS21	47 CS1_21
		CS22	48 CS1_22
		CS23	49 CS1_23
		CS24	50 CS1_24
		CS25	51 CS1_25
		CS26	52 CS1_26
		CS27	53 CS1_27
		CS28	54 CS1_28
		CS29	56 CS1_29
		CS30	57 CS1_30
		CS31	58 CS1_31
		CS32	59 CS1_32
		CS33	60 CS1_33
		CS34	1 CS1_34
		CS35	2 CS1_35
		CS36	3 CS1_36
		CS37	4 CS1_37
		CS38	5 CS1_38
		CS39	6 CS1_39
12	GND		

PVCC 15, PVCC 25, AVCC 14, GND 12

13 x 9 RGB MATRIX: R = CS(3k), G = CS(3k−1), B = CS(3k−2)

CS1_3	D1	D2	D3	D4	D5	D6	D7	D8	D9
CS1_2									
CS1_1									
CS1_6	D10	D11	D12	D13	D14	D15	D16	D17	D18
CS1_5									
CS1_4									
CS1_9	D19	D20	D21	D22	D23	D24	D25	D26	D27
CS1_8									
CS1_7									
CS1_12	D28	D29	D30	D31	D32	D33	D34	D35	D36
CS1_11									
CS1_10									
CS1_15	D37	D38	D39	D40	D41	D42	D43	D44	D45
CS1_14									
CS1_13									
CS1_18	D46	D47	D48	D49	D50	D51	D52	D53	D54
CS1_17									
CS1_16									
CS1_21	D55	D56	D57	D58	D59	D60	D61	D62	D63
CS1_20									
CS1_19									
CS1_24	D64	D65	D66	D67	D68	D69	D70	D71	D72
CS1_23									
CS1_22									
CS1_27	D73	D74	D75	D76	D77	D78	D79	D80	D81
CS1_26									
CS1_25									
CS1_30	D82	D83	D84	D85	D86	D87	D88	D89	D90
CS1_29									
CS1_28									
CS1_33	D91	D92	D93	D94	D95	D96	D97	D98	D99
CS1_32									
CS1_31									
CS1_36	D100	D101	D102	D103	D104	D105	D106	D107	D108
CS1_35									
CS1_34									
CS1_39	D109	D110	D111	D112	D113	D114	D115	D116	D117
CS1_38									
CS1_37									

SW1_1 SW1_2 SW1_3 SW1_4 SW1_5 SW1_6 SW1_7 SW1_8 SW1_9

Drawn by scripts/schematic.py
IS31FL3741A, 13 x 9 RGB. LED references are cell order only
CERN-OHL-P-2.0
Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

Title: round-pendant: driver 1 and matrix

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 4/5

DRIVER 2

Caps at PVCC 15 and 25 and AVCC 14; ground to the chip's GND.
22 uF bulk: edges only. 494 mA scan peak (drivers.md).

VSYS
C40 22u
C41 1u
C42 100n
C43 1u
C44 100n
C45 1u
C46 100n
GND
+3V3
R33 100k
U2
IS31FL3741A
I2C_SDA
I2C_SCL
INTB2
SDB2
C47 100n
R34 100k
R35 30k
SDA 9
SCL 10
INTB 7
SDB 11
PVCC 15
PVCC 25
AVCC 14
ISET 13
ADDR 8
SW1 16 SW2_1
SW2 17 SW2_2
SW3 18 SW2_3
SW4 19 SW2_4
SW5 20 SW2_5
SW6 21 SW2_6
SW7 22 SW2_7
SW8 23 SW2_8
SW9 24 SW2_9
CS1 26 CS2_1
CS2 27 CS2_2
CS3 28 CS2_3
CS4 29 CS2_4
CS5 30 CS2_5
CS6 31 CS2_6
CS7 32 CS2_7
CS8 33 CS2_8
CS9 34 CS2_9
CS10 35 CS2_10
CS11 37 CS2_11
CS12 38 CS2_12
CS13 39 CS2_13
CS14 40 CS2_14
CS15 41 CS2_15
CS16 42 CS2_16
CS17 43 CS2_17
CS18 44 CS2_18
CS19 45 CS2_19
CS20 46 CS2_20
CS21 47 CS2_21
CS22 48 CS2_22
CS23 49 CS2_23
CS24 50 CS2_24
CS25 51 CS2_25
CS26 52 CS2_26
CS27 53 CS2_27
CS28 54 CS2_28
CS29 56 CS2_29
CS30 57 CS2_30
CS31 58 CS2_31
CS32 59 CS2_32
CS33 60 CS2_33
CS34 1 CS2_34
CS35 2 CS2_35
CS36 3 CS2_36
CS37 4 CS2_37
CS38 5 CS2_38
CS39 6 CS2_39
GND 12
GND

ADDR VSYS: 0x33
30 k: 12.67 mA peak,
1.25 mA average per die

13 x 9 RGB MATRIX: R = CS(3k), G = CS(3k−1), B = CS(3k−2)

SW2_1 SW2_2 SW2_3 SW2_4 SW2_5 SW2_6 SW2_7 SW2_8 SW2_9

D118	D119	D120	D121	D122	D123	D124	D125	D126
D127	D128	D129	D130	D131	D132	D133	D134	D135
D136	D137	D138	D139	D140	D141	D142	D143	D144
D145	D146	D147	D148	D149	D150	D151	D152	D153
D154	D155	D156	D157	D158	D159	D160	D161	D162
D163	D164	D165	D166	D167	D168	D169	D170	D171
D172	D173	D174	D175	D176	D177	D178	D179	D180
D181	D182	D183	D184	D185	D186	D187	D188	D189
D190	D191	D192	D193	D194	D195	D196	D197	D198
D199	D200	D201	D202	D203	D204	D205	D206	D207
D208	D209	D210	D211	D212	D213	D214	D215	D216
D217	D218	D219	D220	D221	D222	D223	D224	D225
D226	D227	D228	D229	D230	D231	D232	D233	D234

CS2_3 CS2_2 CS2_1
CS2_6 CS2_5 CS2_4
CS2_9 CS2_8 CS2_7
CS2_12 CS2_11 CS2_10
CS2_15 CS2_14 CS2_13
CS2_18 CS2_17 CS2_16
CS2_21 CS2_20 CS2_19
CS2_24 CS2_23 CS2_22
CS2_27 CS2_26 CS2_25
CS2_30 CS2_29 CS2_28
CS2_33 CS2_32 CS2_31
CS2_36 CS2_35 CS2_34
CS2_39 CS2_38 CS2_37

Drawn by scripts/schematic.py
IS31FL3741A, 13 x 9 RGB. LED references are cell order only
CERN-OHL-P-2.0
Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

Title: round-pendant: driver 2 and matrix

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 5/5